SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks50** **Duration :60 Minutes Annexure A**

Scenario-Based Questions

Code of Conduct:

I B.Purna Kishor Reddy(192472171)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *B.Purna Kishor Reddy*

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Marks (100)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Scenario-Based Questions

1. Low Contrast Image
A grayscale image appears dull with very little difference between object and background intensities. (a) Clearly interpret the scenario and explain the cause of low contrast.
(a) Interpretation of the Scenario
A grayscale image appears dull because the intensity values of the object and the background are very close to each other. As a result, objects are difficult to distinguish from the background.

(b) Identify all key issues affecting image visibility.Small intensity range.
(b) Key Issues Affecting Image Visibility Poor separation between foreground and background.
Loss of important image details.
Reduced visual quality.
Difficult image segmentation.

(c) Apply an appropriate enhancement technique to address the problem

Histogram Equalization

Steps:

1. Read the grayscale image.
2. Calculate the histogram.
3. Compute cumulative distribution function (CDF).
4. Redistribute pixel intensities.
5. Generate enhanced image.

(d) Justify your choice with logical reasoning.

(d) Justification

- Increases image contrast.
- Spreads intensity values over the full range.
- Makes hidden details visible.
- Simple and computationally efficient. ● Suitable for grayscale images.

(e) Present your explanation in a clear and structured manner.

e) Conclusion

Histogram Equalization improves visibility by increasing contrast, making objects easier to detect and analyze.

2. Uneven Illumination
An image captured under non-uniform lighting shows bright and dark patches.
(a) Interpret the scenario and explain the impact of uneven illumination.

(a) Interpretation of the Scenario

The image contains bright and dark regions because lighting is not uniformly distributed across the scene.

(b) Identify key factors causing intensity variation.

(b) Key Factors Causing Intensity Variation

- Non-uniform lighting.
- Shadows.
- Different light source positions.
- Camera exposure variation. ● Surface reflections.

(c) Apply a suitable enhancement method to normalize illumination.

(c) Suitable Enhancement Method

Adaptive Histogram Equalization (CLAHE)

Steps

1. Divide image into small blocks.
2. Perform histogram equalization on each block.
3. Limit excessive contrast.
4. Merge neighboring blocks smoothly.
5. Obtain normalized illumination.

(d) Justify your selected approach with reasoning.

(d) Justification

- Enhances local contrast.
- Corrects uneven brightness.
- Preserves image details.
- Prevents over-enhancement.
- Produces natural-looking images.

(e) Provide a clear and well-structured explanation.

(e) Conclusion

CLAHE effectively normalizes uneven illumination while preserving important image features.

3. Salt-and-Pepper Noise
A scanned document contains salt-and-pepper noise affecting readability.
(a) Explain the scenario and characteristics of this noise.

(a) Interpretation of the Scenario

Salt-and-pepper noise appears as randomly distributed white and black pixels in a scanned document, reducing readability.

(b) Identify key issues impacting image quality.

(b) Key Issues

- Random black pixels.
- Random white pixels.
- Poor readability.
- Loss of character information. ● Reduced OCR accuracy.

(c) Apply an appropriate filtering technique.

(c) Appropriate Filtering Technique

Median Filter

Steps

1. Select window (3×3 or 5×5).
2. Arrange neighboring pixel values.
3. Find median value.
4. Replace center pixel.
5. Repeat for entire image.

(d) Justify why this method is more suitable than alternatives.

(d) Justification

- Removes impulse noise effectively.
- Preserves edges.
- Better than Mean Filter.
- Maintains text quality.

- Produces cleaner documents.

(e) Present your response clearly and logically.

(e) Conclusion

Median filtering removes salt-and-pepper noise while preserving important document details.

4. Blurred Edges
An image processed for noise removal appears overly smooth, causing edges to blur.
(a) Interpret the scenario and explain why edges are blurred.

(a) Interpretation of the Scenario

Noise removal has excessively smoothed the image, causing edges to become blurry and reducing object boundaries.

(b) Identify key issues related to smoothing and edge loss.

(b) Key Issues

- Loss of edge information.
- Reduced sharpness.
- Poor boundary detection.
- Decreased segmentation accuracy. ● Loss of fine details.

(c) Apply an appropriate technique to restore edges.

(c) Appropriate Technique

Unsharp Masking (Image Sharpening)

Steps

1. Blur original image.
2. Subtract blurred image from original.
3. Obtain edge details.
4. Add edge details back.
5. Generate sharpened image.

(d) Justify your decision with clear reasoning.

(d) Justification

- Restores edges.
- Improves sharpness.
- Enhances object boundaries.
- Maintains overall image quality. ● Better than repeated smoothing.

(e) Structure your explanation clearly.

(e) Conclusion

Unsharp Masking restores blurred edges and improves feature visibility for further image analysis.

5. High-Frequency Noise
A satellite image contains fine-grained noise affecting feature visibility.
(a) Interpret the scenario and describe high-frequency noise.

(a) Interpretation of the Scenario

A satellite image contains fine-grained random noise that mainly affects high-frequency components, reducing image clarity.

(b) Identify the key issues affecting image clarity.

(b) Key Issues

- Fine-grained noise.
- Poor feature visibility.
- Reduced image clarity.
- Difficult object detection. ● Lower analysis accuracy.

(c) Apply an appropriate frequency domain filtering method.

(c) Frequency Domain Filtering Method

Low-Pass Filter (Gaussian Low-Pass Filter)

Steps

1. Convert image to frequency domain using FFT.
2. Apply Gaussian Low-Pass Filter.
3. Remove high-frequency noise.
4. Perform inverse FFT.
5. Obtain enhanced image.

(d) Justify the choice of filter with reasoning.

(d) Justification

- Smoothly removes high-frequency noise.
- Preserves important structures.
- Produces fewer artifacts.
- Better than Ideal Low-Pass Filter. ● Suitable for satellite imagery.

(e) Present your answer in a clear and organized manner.

(e) Conclusion

Gaussian Low-Pass Filtering effectively removes high-frequency noise while maintaining important image information.